1.

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY logic_test IS
  PORT (
    a   : IN  STD_LOGIC;
    b   : IN  STD_LOGIC;
    c   : IN  STD_LOGIC;
    d   : IN  STD_LOGIC;
    out1 : OUT STD_LOGIC;
    out2 : OUT STD_LOGIC
  );
END logic_test;

ARCHITECTURE Behavioral OF
logic_test IS
BEGIN

  out1 <= a AND b AND c AND d;

  out2 <= a OR b OR c OR d;

END Behavioral;
```

2.

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY mux8to1 IS
  PORT (
    a   : IN  STD_LOGIC;
    b   : IN  STD_LOGIC;
    c   : IN  STD_LOGIC;
    d   : IN  STD_LOGIC;
    e   : IN  STD_LOGIC;
    f   : IN  STD_LOGIC;
    g   : IN  STD_LOGIC;
    h   : IN  STD_LOGIC;
    ch0 : IN  STD_LOGIC;
    ch1 : IN  STD_LOGIC;
    ch2 : IN  STD_LOGIC;
    data : OUT STD_LOGIC
  );
END mux8to1;

ARCHITECTURE Behavioral OF mux8to1 IS
BEGIN

  data <= a WHEN ch2 = '0' AND ch1 = '0' AND
ch0 = '0' ELSE
       b WHEN ch2 = '0' AND ch1 = '0' AND ch0
= '1' ELSE
       c WHEN ch2 = '0' AND ch1 = '1' AND ch0
= '0' ELSE
       d WHEN ch2 = '0' AND ch1 = '1' AND ch0
= '1' ELSE
       e WHEN ch2 = '1' AND ch1 = '0' AND ch0
= '0' ELSE
       f WHEN ch2 = '1' AND ch1 = '0' AND ch0
= '1' ELSE
       g WHEN ch2 = '1' AND ch1 = '1' AND ch0
= '0' ELSE
       h WHEN ch2 = '1' AND ch1 = '1' AND ch0
= '1' ELSE
       '0';

END Behavioral;
```

3.

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY display_letters IS
  PORT (
    sel : IN  STD_LOGIC_VECTOR(2 DOWNTO 0);
    seg : OUT STD_LOGIC_VECTOR(7 DOWNTO
0)
  );
END display_letters;

ARCHITECTURE Behavioral OF display_letters
IS
BEGIN

  -- Segment order: a, b, c, d, e, f, g, dp
  -- '1' = ON, '0' = OFF

  seg <= "11101110" WHEN sel = "000" ELSE  --
a
       "00111110" WHEN sel = "001" ELSE  -- b
       "00011010" WHEN sel = "010" ELSE  -- c
       "01111010" WHEN sel = "011" ELSE  -- d
       "10001110" WHEN sel = "100" ELSE  -- f
       "01110110" WHEN sel = "101" ELSE  -- g
       "00000000";

END Behavioral;
```

4.

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY voting25 IS
  PORT (
    votes  : IN  STD_LOGIC_VECTOR(24 DOWNTO
0);
    result : OUT STD_LOGIC
  );
END voting25;

ARCHITECTURE Behavioral OF voting25 IS
BEGIN

  PROCESS(votes)
    VARIABLE count : INTEGER RANGE 0 TO 25;
  BEGIN
    count := 0;

    FOR i IN votes'RANGE LOOP
      IF votes(i) = '1' THEN
        count := count + 1;
      END IF;
    END LOOP;

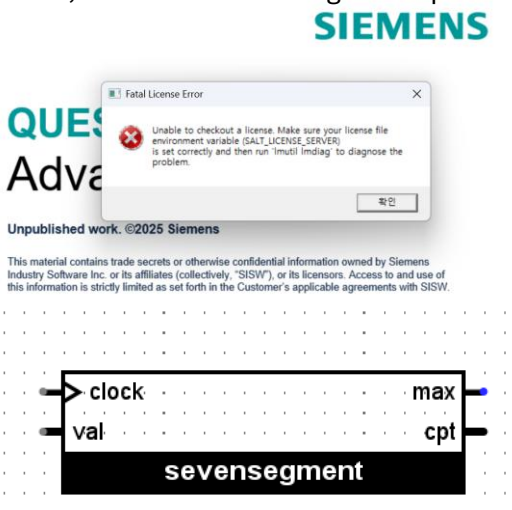
    IF count >= 13 THEN
      result <= '1';
    ELSE
      result <= '0';
    END IF;

  END PROCESS;

END Behavioral;
```

Result.

I can’t use questa
I don't know why this doesn’t work
Also, VHDL doesn’t change it’s input and output.



5.

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY count_ones IS
  PORT (
    en  : IN  STD_LOGIC;
    vt  : IN  STD_LOGIC_VECTOR(12 DOWNTO 0);
    seg : OUT STD_LOGIC_VECTOR(6 DOWNTO 0)
  );
END count_ones;

ARCHITECTURE Behavioral OF count_ones IS
BEGIN

  PROCESS(en, vt)
    VARIABLE count : INTEGER RANGE 0 TO 13;
  BEGIN

    IF en = '0' THEN
      seg <= "0000000";

    ELSE
      count := 0;

      FOR i IN vt'RANGE LOOP
        IF vt(i) = '1' THEN
          count := count + 1;
        END IF;
      END LOOP;

      CASE count IS
        WHEN 0  => seg <= "1111110";
        WHEN 1  => seg <= "0110000";
        WHEN 2  => seg <= "1101101";
        WHEN 3  => seg <= "1111001";
        WHEN 4  => seg <= "0110011";
        WHEN 5  => seg <= "1011011";
        WHEN 6  => seg <= "1011111";
        WHEN 7  => seg <= "1110000";
        WHEN 8  => seg <= "1111111";
        WHEN 9  => seg <= "1111011";
        WHEN 10 => seg <= "1110111"; -- A
        WHEN 11 => seg <= "0011111"; -- b
        WHEN 12 => seg <= "1001110"; -- C
        WHEN 13 => seg <= "0111101"; -- d
        WHEN OTHERS => seg <= "0000000";
      END CASE;

    END IF;

  END PROCESS;

END Behavioral;
```